QCoDeS-Based Cryogenic Transport Measurement Platform for Quantum Dot Devices

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Introduction

Single-electron transistors (SETs) are nanoscale devices that exploit Coulomb blockade to control the transport of individual electrons. These devices serve as a key platform for studying semiconductor quantum dots and are closely related to architectures used in quantum computing. Quantum device structures are fabricated using the NanoFrazor thermal scanning probe lithography system.

We report on the development of a cryogenic electronic transport spectroscopy platform for semiconductor quantum dot devices. The setup integrates low-noise source meters and signal generators with computer-controlled gate electrodes to enable flexible biasing and pulsed-gate experiments. Instrument control and data acquisition are implemented using the QCoDeS Python-based framework, providing a modular interface for adding new instruments and designing complex multi-parameter sweeps. This architecture facilitates automated measurements, reproducible configuration of experiments, and streamlined data management for future studies of quantum coherence and spin-dependent transport.

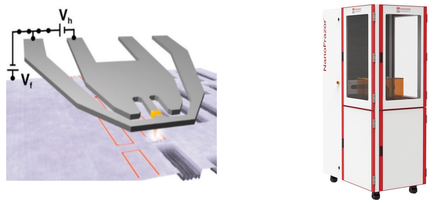

Fig 1. The NanoFrazor lithography system [1]

Problem Statement and Objectives

Problem Statement

Semiconductor quantum devices require precise electrical measurements at cryogenic temperatures while controlling multiple instruments and parameters. Traditional manual measurement methods are inefficient, difficult to reproduce, and prone to errors. Therefore, an automated, software-controlled framework is needed to enable reliable measurements, scalable experimentation, and efficient data management for quantum device research.

Objectives

- Implement a measurement framework using QCoDeS for automated electrical characterization.
- Perform traditional I–V curve measurements on a known device (MOSFET) to validate the measurement system.
- Demonstrate scalable experimental automation applicable to semiconductor quantum dot and qubit devices.

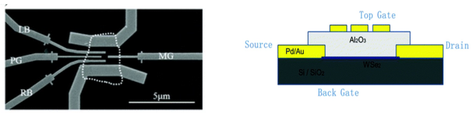

Fig 2. Lithographically defined Quantum Dot Device[3]

Methodologies

Software Stack

QCoDeS is a Python-based data acquisition framework developed by the Copenhagen / Delft / Sydney / Microsoft quantum computing consortium[2]. Some of the reasons behind using QCoDeS are identified below:

1. It is an open-source, Python-native, and scriptable framework that follows a well-defined object-oriented abstraction model.
2. It is reproducible across teams and has broad support for most instrument drivers, with active community maintenance.
3. It offers a better alternative to GUI-based approaches by providing finer control over measurements, which is essential in specialized applications.

Hardware

A Keithley 2400 SourceMeter was used to apply and measure the drain–source voltage and current of the device under test, while a Rigol DG1022Z Pro Waveform Generator provided the gate voltage.

Working:

Custom Python scripts automated instrument control, voltage sweeps, and data acquisition. Measurement results, including I–V curves, were visualized, enabling real-time plotting and structured data management for device characterization. NumPy arrays were used to save the results for further analysis and future references.

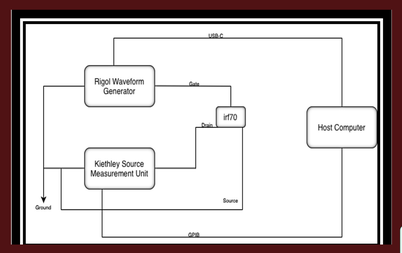

Fig 3. Wiring Schematic of the setup.

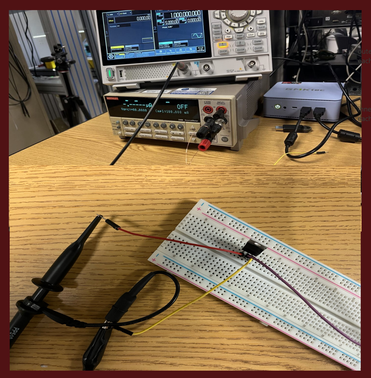

Fig 4. Laboratory Setup with the SourceMeter and the Waveform Generator

Fig 5. Flowchart of the code

Research Progress and Result

- The characterization methodology was validated successfully across all standard devices.
- Output IV curves for the IRF510 power MOSFET match the manufacturer datasheet, confirming the measurement setup.
- The QCoDeS-based automation pipeline is working correctly.

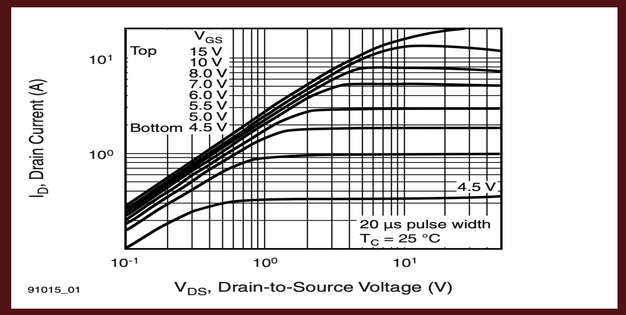

Fig 6. The theoretical I_DV_{DS} curve

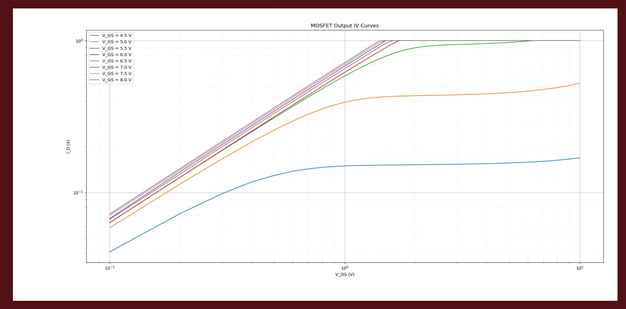

Fig 7. The experimental I_DV_{DS} curve

Potential Challenges

•At cryogenic temperatures, parasitic resistances in wiring and contact pads can dominate over the device signal.

•As device dimensions shrink to quantum dot scale, gate oxide integrity becomes harder to maintain, risking leakage currents that obscure measurement.

•Unlike MOSFETs, quantum dot devices have discrete energy levels that make sweeping V_{GS} less predictable without prior spectroscopy.

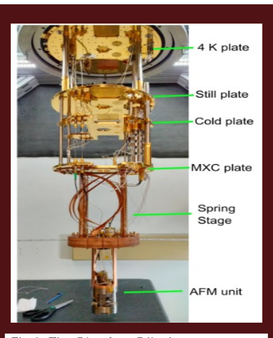

Fig 8. The Bluefors Dilution Refrigerator at Texas State University

Conclusion and Future Work

An automated electronic measurement platform was developed using the QCoDeS framework to control instruments and perform device characterization. The system successfully integrates a Keithley 2400 SourceMeter and a Rigol DG1022Z Pro Waveform Generator to conduct automated voltage sweeps and acquire electrical transport data. Measurement results were visualized in an I_D V_{GS} plot, enabling efficient data monitoring and analysis. The platform demonstrates a scalable and reproducible workflow for electronic transport measurements applicable to semiconductor and quantum device experiments.

Future work will focus on extending the platform to characterize nanoscale quantum devices such as single-electron transistors and quantum dots. Additional measurement capabilities, including multi-parameter sweeps and pulsed-gate experiments, will be implemented to study quantum transport phenomena.

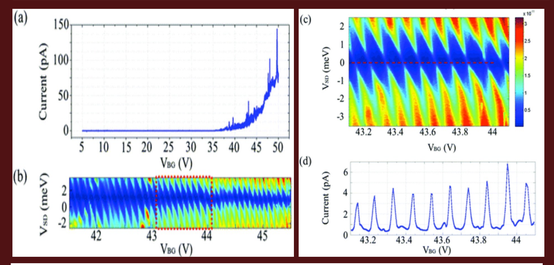

Fig 9. (a) I_D vs V_{BG}, (b) V_{SD} vs V_{BG} showing Coulomb diamonds, (c) V_{SG} vs V_{BG}, (d) I_D vs V_{BG} showing Coulomb peaks. [3]

References

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